Special Topics in Physics (ASTR 6991 / FISI 4997): Astrophysics
Mini-Project #2
Due: Friday, May 22, 2026 (5pm)

Student Name: ______________________________

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Introduction

The goal of this exercise is to create your own rotation curve of our galaxy, the Milky Way. A good way to start with this is to collect data of the 21 cm across the galaxy. Fortunately, you can access online archival data of the 21cm spectral line, at `https://www.astro.uni-bonn.de/hisurvey/AllSky_gauss/`. The following sections will guide you through the process of collecting and analyzing your data, and ultimately creating your rotation curve. *Please record all of your data, assumptions, and conclusions along the way and include them in your report in a reproducible fashion. No two students should have the same positions, but please feel free to work together. If your discussion with another student impacts your project in any way, cite that student where appropriate. Required constants (galactocentric radius and rotation speed, e.g.) should be taken from the textbook.*

Part 1: The Inner Galaxy

In principle, you could look for HI spectra in any direction. For this exercise, it's best to stick close to the plane (think b latitude $+/-$ a few degrees). Recall that a rotation curve requires plotting rotational velocity vs. distance. The velocity you get from the spectra, however, will be the radial velocity. Fortunately, you can use a method in your book (see Chapter 24) to obtain rotational velocities from your measurements.

Choose a range of longitudes across the inner galaxy (at least 8 positions). Visit the website given in the intro to obtain HI4PI HI spectra for your desired longitudes.

Enter the l, b (galactic coordinates). Choose a small beam size (similar to a field of view) to minimize the complication of your spectra. Finally, leave the velocity range set on the default. What range of radial velocities should you expect to detect if you look in all regions of the galaxy? Reminder: record all parameters used to obtain your spectra in the interface, as well as screenshots of your spectra (for reference), gaussian plots, and gaussian parameters.

After you enter the parameters, click search. Take the screenshot of your spectrum, and also save it as a text file by clicking "Download the HI spectra as ASCII file."

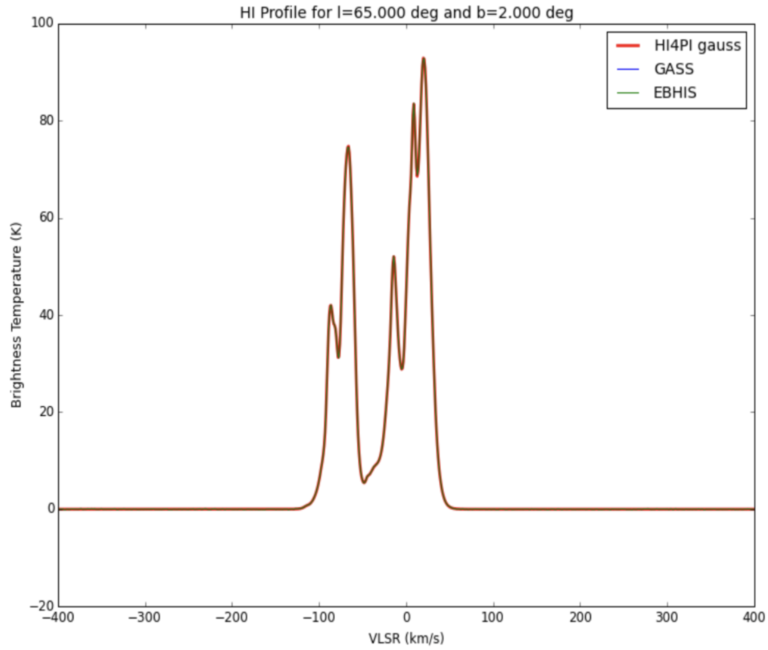
How do you know when you've got a useful spectrum? Try to choose regions that give you clean HI spectra, where the peaks are easily separable. You will see the difference if you start jumping around to various positions. See the example below. It has multiple peaks, but two major components to the distribution. The gas near velocity 0 km/s is likely local gas, and it won't help much with our exercise. However, the gas near -90 km/s is representing more distance gas, and this is useful for our purpose. If you find a spectrum where you can't easily separate the gas populations, try to choose a different region.

Part 2: Fitting your spectra

After you have your .txt file for each position, use the attached python program (if you like) to fit gaussian models to your data. You need to do this for each spectrum, but after you get used to it this will only take a few minutes. The gaussian models give you three specific parameters - the central velocity of the spectral line with respect to the LSR, the amplitude of the spectral line, and the full width at half maximum linewidth. For this project, we will use the Vlsr. This parameter will be printed in a separate file for each position you fit (you need to remember to change your filename for each position, else you'll overwrite it each time you run it).

The program will ask you a series of questions, and instructions for how to proceed are below.

1. When your code runs, you will see a plot window with a spectrum. The program will ask you to specify an x-range - this just means select (with your mouse) the left and right points you want to keep along the horizontal axis. You can click on the far left and click on the far right systematically, and that will suffice.
2. Enter the number of gaussian components. If you have a spectrum like the example shown here, it is fine to



focus on the gas at -90 km/s. You can completely ignore the gas at 0 km/s in this case (you do not need to fit it).
3. Select the top and bottom point for Gaussian amplitude. You will click at the top of your peak(s), and click at the bottom (near 0 on the vertical axis).
4. Select the center of the gaussian. Click on the horizontal axis where you think is the center of your gas distribution (near -90 km/s for the example). 5. Select the full width at half maximum. This is the width of the spectral line at a heigh of half of the maximum amplitude. In our example, the height is around 80 K for the largest peak. At 40 K, we would click once on the left, and once on the right side of the spectral line.
6. After each step, the software will ask you if you want to add any constraints. You can enter 0 every time.
7. After you confirm the gaussian fit (making sure the shape of the model roughly fits with your data, aligning with the LSR velocity, most importantly), hit y, and the program will save a .txt file with your gaussian parameters. The central velocity can be used for your calculations.

Note: The difference between Gaussian centers and terminal velocities is typically small (a few km/s), but can introduce a systematic bias. You may use Gaussian fits as a guide, but be aware that the true terminal velocity lies at the edge of the emission profile. If your HI spectra are very wide or extended in the region of the fastest moving emission, you may need to use the terminal velocity instead of the peak (central or Vlsr). This is up to your discretion (so, explain!).

Part 3: The Outer Galaxy

The method you used so far only works for a certain range of longitudes. What is that range, and why?

If we want to continue our rotation curve to large distances, we may need to devise a way to use measurements from longitudes representing the outer galaxy. However, this usually requires knowing the distance by using standard candles, and ain't nobody got time for that! So, we are going to rest on the hard work of others and try using sources with known distances.

The WISE survey covers the galactic plane and includes a catalog of HII regions (https://astro.phys.wvu.edu/wise/). These regions can be searched across the longitude range of interest and selected for known distances and even radial velocities. Be sure to pick regions maximizing the range of *galactocentric* distances.

Make sure to also pick a few regions in the inner galaxy, just so you can compare your results with those of the first method you used.

Side note: Could you have just used this catalog procedure for the inner galaxy? (Yes, but it wouldn't have been as much fun).

Part 4: Analysis

Please include the following in your analysis. You are NOT limited to discussing only this - all detailed research findings and methodologies should be included in the report (even if I didn't explicitly ask about it!).

1. To create your final plot, please include both your inner galaxy and outer galaxy data points (with some overlap between the longitude ranges). Once you have both plotted, overplot the rotation curve we would expect for a Keplerian model. Is there a discrepancy? Discuss to what extent they differ or diverge, and give an explanation.

2. In any series of measurements, uncertainties arise from the measurement techniques, assumptions, or other biases. Can you quantify your uncertainties for this exercise? Include any possible error bars on your plot as well as a discussion of additional sources of uncertainty.

3. Fortunately, there are existing results in the literature to which we can compare our experiment[1]. Using either the references in the text or, e.g., the results from Sofue 2013, overplot the data reported in the paper in your plot. Discuss how it compares with your results, and include your assessment of any discrepancies.

4. Finally, could you conduct this experiment for other galaxies? If so, how would the setup differ, if at all, for spiral vs. elliptical galaxies? What would you need to change about the setup of our HI4PI queries for spiral galaxies? What other data sets would you use (if any) for ellipticals?

References

Sofue, Y. 2013, PASJ, 65, 118

[1] Feel free to use any reference you like. Just be sure you check the constants used and can give a 1-2 sentence summary of the method used to get their rotation curve.